

Arthroscopically Assisted Reconstruction of the Anterior Cruciate Ligament

A PROSPECTIVE RANDOMIZED ANALYSIS OF THREE TECHNIQUES*

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ABSTRACT: One hundred and twenty-seven patients who had a rupture of the anterior cruciate ligament agreed to participate in a prospective, randomized study of three arthroscopically assisted reconstruction techniques. One hundred and twenty-five patients (125 reconstructions) were evaluated after a mean duration of follow-up of forty-two months (range, two to five years).

Group I included forty patients who had a two-incision reconstruction with use of an autogenous semitendinosus-gracilis graft, group II consisted of forty patients who had a two-incision reconstruction with use of an autogenous patellar-ligament graft, and group III included forty-five patients who had a single-incision reconstruction (endoscopic technique) with use of an autogenous patellar-ligament graft. The male-female ratio, age range, level of athletic activity, interval between the injury and the reconstruction, previous operative procedures, and associated injuries were similar in all three groups. The same postoperative rehabilitation protocol was followed for all patients.

Testing with a KT-2000 arthrometer at maximum manual force was done at the follow-up evaluation; the difference in laxity between the involved knee and the contralateral knee was three millimeters or less in thirty-three patients (83 per cent) in group I, thirty-seven patients (93 per cent) in group II, and thirty-nine patients (87 per cent) in group III. A difference of two millimeters or less was found in thirty patients (75 per cent) in group I, thirty-one patients (78 per cent) in group II, and thirty-five patients (78 per cent) in group III. Thirty-five patients (88 per cent) in group I, thirty-eight patients (95 per cent) in group II, and forty patients (89 per cent) in group III returned to at

least the same level of athletic activity. Four grafts (two in group I and two in group II) failed as a result of trauma. There was one additional failure in groups I and III, as evidenced by a difference of nine and seven millimeters, respectively, on instrumented testing of laxity.

The significant findings were that no knee was rated D according to the system of the International Knee Documentation Committee ($p < 0.002$, 94 per cent confidence level) and that fewer additional operative procedures were done on patients in group III ($p < 0.08$). Also, it was found that the patients in group II returned to a greater level of athletic activity ($p < 0.02$) and that a higher percentage of the patients in this group had a difference of three millimeters or less on testing with the KT-2000 arthrometer than in the other two groups ($p < 0.08$).

However, with the numbers available, there were no significant differences in the over-all outcome among the three groups ($p > 0.1$). Importantly, the rate of failure was not greater and the outcomes were not less satisfactory for the late reconstructions than they were for the acute reconstructions (those performed less than three weeks after the injury), including those done with an autogenous semitendinosus-gracilis graft in a chronically unstable knee.

A rupture of the anterior cruciate ligament compromises the stability of the knee in individuals who are active^{2,6,10-13,20,22,40,41}, resulting in instability, recurrent injury, and associated intra-articular disease^{2,5,10-13,22,29}.

Both open and arthroscopically assisted techniques for reconstruction of the anterior cruciate ligament with a graft have successfully restored the stability of the knee^{3,6,15,16,18,20,23,24,31,35,36,40-42}. Stability has also been reported with several different arthroscopically assisted techniques^{3,14,19,34} with a variety of autogenous grafts for the reconstruction^{1,27,30}. The two most commonly used autogenous grafts are the semitendinosus-gracilis medial-hamstring-tendon construct and the central one-third of the patellar ligament. Disadvantages of the former include a failure to achieve immediate rigid fixation to bone^{35,38} and decreased stiffness as compared with a patellar ligament graft or the native cruciate ligament^{20,29,38}.

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Disadvantages of the latter include patellofemoral pain, weakness of the quadriceps muscle, and, rarely, rupture of the patellar ligament or patellar fracture^{20,33,36,40}.

Some authors have expressed concern about the use of rapid rehabilitation programs after reconstruction with a semitendinosus-gracilis graft because of the elasticity of the graft^{1,20,21,24,27,30,32,38,40}. Concern has also been expressed about the non-optimum fixation of patellar ligament-bone blocks that may result from divergence of the femoral interference screw, which is difficult to place during arthroscopic reconstruction with use of the single-incision technique^{19,23,25}.

The purpose of this study was to determine if there was a difference in the outcome among three operative techniques: arthroscopically assisted (two-incision) reconstruction of the anterior cruciate ligament with a semitendinosus-gracilis graft, the same procedure with a patellar ligament graft, and a single-incision (endoscopic) reconstruction with a patellar ligament graft.

Materials and Methods

Two hundred and thirty-six patients who had a tear of the anterior cruciate ligament were examined between August 1989 and August 1992. After counseling, seventy-two individuals chose to modify their activity and to be managed non-operatively. Thirty-five patients who had decided to have a reconstruction declined to be included in the study.

The study groups comprised the remaining 129 patients. Each patient was informed that, although all three procedures had had proved clinically successful outcomes, success could not be guaranteed and failures as well as complications had been reported for each technique. Institutionally approved informed consent to enter the study was then obtained. The patients were randomly assigned to the type of operative procedure according to the month in which they had been born. Group I (two-incision arthroscopically assisted reconstruction with a semitendinosus-gracilis graft) included patients whose birthday was in January through April; group II (two-incision arthroscopically assisted reconstruction with a patellar ligament graft), May through August; and group III (single-incision endoscopic reconstruction with a patellar ligament graft), September through December. No exceptions were made for patients who had had a failure of a previous operation on the anterior cruciate ligament. Thus, two patients who had had a failed patellar-ligament reconstruction were ineligible for the study because, on the basis of their birthday, they would have to have been assigned to a reconstruction with a patellar ligament graft. One patient in group II and one in group III were lost to follow-up. Thus, 125 patients who had been followed for two to five years (mean, forty-two months) were evaluated for the study.

Group I consisted of forty patients (forty operations), who were a mean of twenty-seven years old

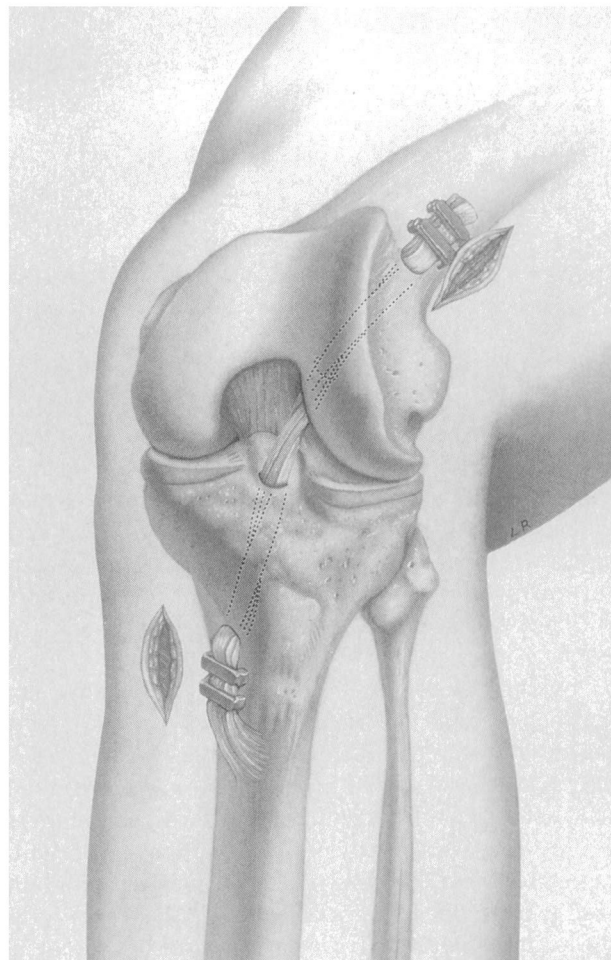


FIG. 1-A

Figs. 1-A, 1-B, and 1-C: The two-incision arthroscopically assisted reconstruction of the anterior cruciate ligament with an autogenous semitendinosus-gracilis graft.

Fig. 1-A: Artist's rendering of the reconstruction. The graft is secured with barbed staples.

(range, fourteen to fifty-six years old); twenty-seven were male and thirteen were female. Group II included forty patients (forty operations), who were a mean of twenty-six years old (range, fifteen to forty-nine years old); twenty-six were male and fourteen were female. Group III consisted of forty-five patients (forty-five operations), who were a mean of twenty-eight years old (range, fourteen to fifty-six years old); twenty-eight were male and seventeen were female.

Ninety-four patients (75 per cent) — twenty-nine (73 per cent) in group I, thirty-one (78 per cent) in group II, and thirty-four (76 per cent) in group III — had the operation within a year after the initial injury. The operation was done within three weeks after the initial injury in eighteen patients (45 per cent) in group I, seventeen (43 per cent) in group II, and seventeen (38 per cent) in group III. All of the patients were followed for a minimum of two years (range, two to five years). The mean duration of follow-up was thirty-eight months (range, twenty-four to fifty-three months) for group I, forty-eight months (range, twenty-four to sixty months)



FIG. 1-B



FIG. 1-C

Anteroposterior and lateral radiographs of the knee, made at forty months. The patient was an eighteen-year-old woman who played collegiate basketball.

for group II, and thirty-nine months (range, twenty-four to fifty-six months) for group III.

Operative Technique

At the operation, all of the knees were first examined for associated instability, while the patient was under anesthesia. An arthroscopic examination was then done, and any associated injuries were treated. Injuries of the medial collateral ligament and the medial complex were repaired operatively in the first year of this study and were treated non-operatively thereafter. I performed all of the procedures.

Two-Incision Technique with a Semitendinosus-Gracilis Graft

The pes anserinus is identified through a four-centimeter vertical incision, beginning four centimeters distal to the anteromedial tibial joint line (Figs. 1-A, 1-B, and 1-C). The semitendinosus and gracilis tendons are obtained with a tendon-stripper (Orthopedic Systems, Hayward, California) and left attached distally. A four-centimeter vertical incision is then made laterally, superior to the lateral femoral condyle, and the vastus lateralis is elevated anteriorly from the femur. A generous notchplasty is performed through the arthroscope, and drill-guides (Acufex Microsurgical, Norwood, Massachusetts) are used to locate isometric points on the femoral roof and the tibia. A tensiometer (Acufex Microsurgical) was used only for the first several patients in each of the three groups. Tunnels, eight millimeters in diameter, are then created in the femur and the tibia, and the semitendinosus and gracilis tendons

for grafting are passed, under arthroscopic visualization, in a retrograde direction with use of a passing suture. The grafts are secured to the anteromedial aspect of the proximal portion of the tibia with use of two medium barbed staples (Smith and Nephew Richards, Memphis, Tennessee). Pre-tensioning of the graft is accomplished by moving the knee through twenty-five repetitions of flexion and extension; the grafts are then secured to the lateral femoral cortex with two small barbed staples while the knee is in 10 degrees of flexion, to achieve greatest stability at nearly full extension. No posterior force is applied to the tibia.

Two-Incision Technique with a Patellar Ligament Graft

The central one-third of the patellar ligament is obtained through a six-centimeter incision, made from the inferior patellar pole to the medial aspect of the tibial tubercle (Figs. 2-A, 2-B, and 2-C). The ligament is taken in continuity with bone blocks from the inferior, extra-articular aspect of the patella and the tibial tubercle. The blocks are one centimeter wide, two centimeters long, and six millimeters deep. They are obtained with use of an oscillating saw and a one-quarter inch (0.64-millimeter) curved osteotome. Two sutures are passed through drill-holes in each bone block.

Tunnels, ten millimeters in diameter, are created at isometric points in the femur and tibia with the same technique as is used with a semitendinosus-gracilis graft. A passing suture is used to pass the graft retrograde under arthroscopic visualization. The proximal bone block is secured to the femur with a nine by twenty-five-

TABLE I
CONCOMITANT OPERATIVE PROCEDURES*

	Semi- tendinosus- Gracilis Graft (Group I)	Patellar Ligament Graft	
		Two-Incision Technique (Group II)	Endoscopic Technique (Group III)
Repair of medial meniscus	7	4	7
Partial medial meniscectomy	7	5	11
Repair of lateral meniscus	3	2	6
Partial lateral meniscectomy	11	9	14
Repair of medial collateral ligament†	4	0	2
Reconstruction of posterior cruciate ligament	2	0	0
Removal of loose osteochondral body	4	7	1
Abrasion chondroplasty	1	0	1
Excision of meniscal cyst	0	0	1

*The values are given as the number of procedures.

†An additional eleven torn medial collateral ligaments (three in group I, three in group II, and five in group III) were not repaired.

millimeter interference screw (DePuy, Warsaw, Indiana). Pre-tensioning of the graft is accomplished by moving the knee through twenty-five repetitions of flexion and extension, and the distal bone block is secured to the tibia with a similarly sized interference screw while the knee is in 10 degrees of flexion.

Single-Incision (Endoscopic) Technique with a Patellar Ligament Graft

The autogenous patellar-ligament graft is obtained with the same technique as is used with the two-incision procedure. A ten-millimeter-diameter tunnel is made in the tibia, and the isometric point on the posterior aspect of the femoral intercondylar roof is located endoscopically through the tibial tunnel, with use of a fixed-offset guide, seven millimeters anterior to the posterior wall of the notch. The femoral guide-wire is passed to this point through the tibial tunnel, and a cannulated ten-millimeter drill-bit is passed over the wire to create the femoral tunnel.

After the interior surfaces of the tunnels have been smoothed, the patellar ligament graft is passed in retrograde fashion through the tibial tunnel and the proximal bone block and is introduced into the femoral tunnel under arthroscopic guidance (Figs. 3-A, 3-B, and 3-C). A guide-wire is placed parallel and is used to direct the fixation of a nine by twenty-five-millimeter cannulated interference screw into the femur. The guide-wire and the screw are placed through the anterior fat pad just medial to the border of the patellar ligament. Pre-tensioning of the graft is then accomplished by applying tension to the distal bone block, through two controlling

sutures, while the knee is moved through twenty-five repetitions of flexion and extension. After it has been determined that there is no impingement of the graft and that the graft is isometric, the distal tibial bone block is secured with a similar sized interference screw, with the knee in 10 degrees of flexion. In seventeen patients, long patellar-ligament grafts were secured to the proximal aspect of the tibia with two barbed staples.

Previous Operations

Eleven patients in group I had had sixteen previous operative procedures on the knee. Nine patients had had one operative procedure, one had had two, and one had had five. Previous operations involving the anterior cruciate ligament included two reconstructions and three débridements.

Eleven patients in group II had had thirteen previous operative procedures on the knee. Nine patients had had one operation, and two patients had had two. The operations included three reconstructions and

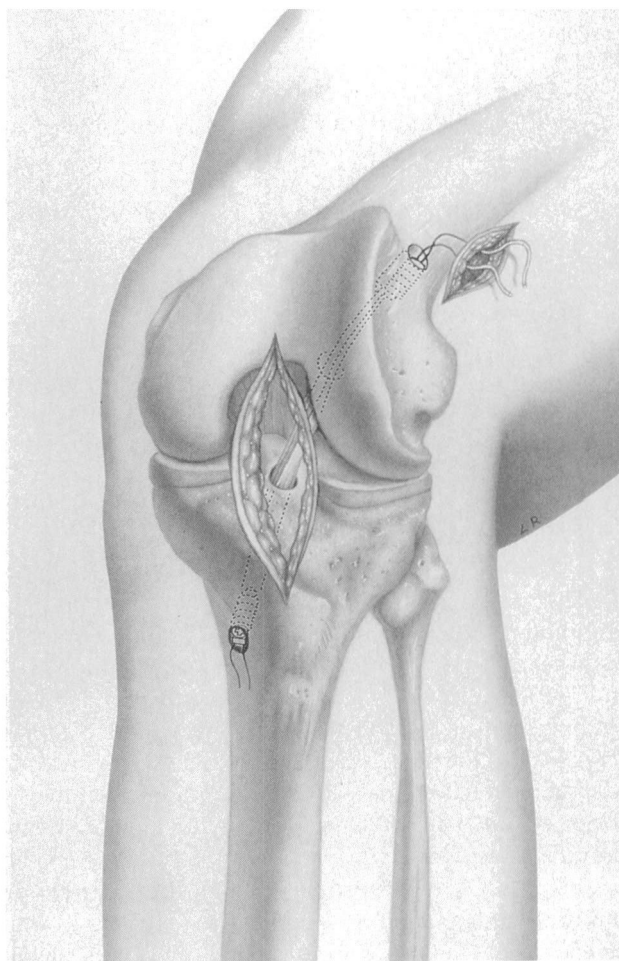


FIG. 2-A

Figs. 2-A, 2-B, and 2-C: The two-incision arthroscopically assisted reconstruction of the anterior cruciate ligament with an autogenous patellar-ligament graft.

Fig. 2-A: Artist's rendering of the reconstruction. The graft is secured with interference screws.

three débridements of the anterior cruciate ligament. One extra-articular reconstruction had been done.

In group III, eighteen patients had had eighteen previous operations on the knee. The operations on the anterior cruciate ligament included two reconstructions and four débridements. Three extra-articular reconstructions had been done.

Concomitant Operations

Twenty-eight (70 per cent) of the patients in group I had at least one additional injury in the ipsilateral knee. Concomitant with the reconstruction of the anterior cruciate ligament, seven medial and three lateral menisci were repaired and seven medial and eleven partial lateral meniscectomies were performed (Table I). Loose osteochondral bodies were removed from four knees, and four medial collateral ligaments were repaired. Two reconstructions of the posterior cruciate ligament and one abrasion chondroplasty were performed. Three torn medial collateral ligaments and two partially torn posterior cruciate ligaments were not repaired.

Twenty-four (60 per cent) of the patients in group II had at least one additional injury in the ipsilateral knee (Table I). At the time of the index reconstruction, four medial and two lateral menisci were repaired and five medial and nine partial lateral meniscectomies were performed. Loose osteochondral bodies were excised from seven knees. A tear was identified in three medial collateral ligaments, but none of them was repaired.

Thirty-five (78 per cent) of the patients in group III had at least one additional injury in the ipsilateral knee

(Table I). Seven medial and six lateral menisci were repaired, and eleven medial and fourteen partial lateral meniscectomies were done. Two of seven torn medial collateral ligaments were repaired. An osteochondral fragment and a meniscal cyst were excised from one knee each, and an abrasion chondroplasty was done in one knee. A tear was found in one lateral collateral ligament, but it was not repaired.

Postoperative Regimen

The patients in all three groups stayed in the hospital overnight and were discharged the next morning. The postoperative regimen did not vary among the groups or if a concomitant operation had been done. The patients were asked to begin full weight-bearing the morning after the operation and were allowed full motion of the knee. The patients were instructed regarding resistance exercises for the hamstrings before they were discharged from the hospital, but closed-kinetic-chain progressive-resistance exercises for the quadriceps were delayed for six weeks³⁶. No crutches or assistive devices were allowed after the first postoperative week.

Running was permitted at three months, but pivoting and cutting activities were not allowed until after the fourth postoperative month. Jumping, and hopping with the involved limb, were not permitted until after the fifth postoperative month. The patient was allowed to return to competitive athletic activity after the sixth postoperative month.

Isokinetic testing of the extensors and flexors of the knee was done with a dynamometer (Biodex, Shirley, New York) at eighteen weeks and six months postop-



FIG. 2-B



FIG. 2-C

Anteroposterior and lateral radiographs of the knee, made at forty-eight months. The patient was a twenty-one-year-old man who played collegiate football as a defensive back.

eratively and at yearly intervals after the index procedure. Tests were performed at 60, 180, 240, and 300 degrees per second. Deficits were recorded at all of the levels tested.

Instrumented testing of laxity was performed with the KT-2000 arthrometer (MedMetric, San Diego, California) six months postoperatively and at yearly intervals after the operation^{7,9}. The quadriceps active test⁷ was used to determine the neutral position of the knee in patients who had posterior laxity.

Orthopaedic evaluations were performed at one, six, and twelve weeks, at six months, and annually thereafter. Radiographs were made intraoperatively, at one and six weeks and then on an annual basis. At yearly intervals, function was evaluated according to the systems of Lysholm and Gillquist²⁶, as modified by Tegner and Lysholm³⁹, and the International Knee Documentation Committee¹⁷, as well as with the one-leg-hop for distance test⁸. All of the instrumented tests and scoring evaluations were done by the same independent physical therapist, to enhance reproducibility.

Results

Timing of the Operation

With the numbers available, there was no significant difference in the outcomes among the three groups or within a group that could be related to the duration of time from the injury to the operative reconstruction ($p < 0.1$).

Return to Pre-Injury Level of Activity

Thirty-five (88 per cent) of the forty patients in group I, thirty-eight (95 per cent) of the forty patients in group II, and forty (89 per cent) of the forty-five patients in group III were able to return to at least the pre-injury level of athletic activity. The patients in group II returned to a significantly greater level of activity than the patients in the other two groups ($p < 0.02$, Student *t* test, 90 per cent confidence level).

At the most recent follow-up evaluation, eighty-one (65 per cent) of the entire study population participated in sports that involved cutting, pivoting, and contact. There was no difference in the level of involvement in the sport or the amount of inherent risk of the sport among the three groups.

Failure of the Graft

There were two traumatic failures of the graft in group I and two in group II.

In group I, the graft in a sixteen-year-old girl failed, twenty-one months postoperatively, while she was executing a non-contact cutting maneuver during a soccer game. The other graft failed four years postoperatively in a thirty-nine-year-old woman, after a skiing accident involving hyperextension of the knee. Both patients had a second reconstruction with an autogenous patellar-ligament graft.

In group II, one failure occurred, fifteen months postoperatively, as a result of a valgus blow to the knee in a seventeen-year-old football player. A second reconstruction was done with use of a semitendinosus-gracilis graft. At the time of the latest follow-up, the patient was playing football for a team in Division I of the National Collegiate Athletic Association. The other failure in this group occurred forty-five months postoperatively in a twenty-five-year-old man who fell sixteen feet (4.9 meters) from a ladder and landed with the knee hyperextended. He also had a second reconstruction, performed with a semitendinosus-gracilis graft.

Testing with the KT-2000 Arthrometer

On testing with the KT-2000 arthrometer at maximum manual force, one patient (group I) had a side-to-side difference in laxity of nine millimeters and one patient (group III) had a difference of seven millimeters. However, both of these patients were asymptomatic, fully participated in sports, and gave the knee a score of

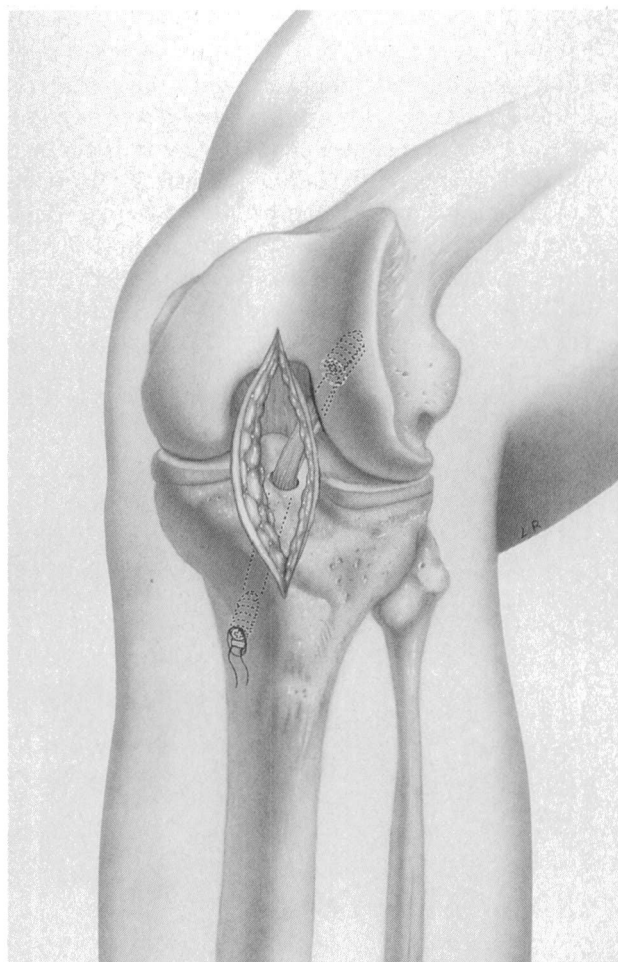


FIG. 3-A

Figs. 3-A, 3-B, and 3-C: The single-incision (endoscopic) reconstruction of the anterior cruciate ligament with an autogenous patellar-ligament graft.

Fig. 3-A: Artist's rendering of the reconstruction. The graft is secured with the use of interference screws; the femoral screw is placed arthroscopically.

TABLE II
POSTOPERATIVE STABILITY

Side-to-Side Difference on Instrumented Testing of Laxity (mm)	Semitendinosus- Gracilis Graft (Group I)*	Patellar Ligament Graft*	
		Two-Incision Technique (Group II)	Endoscopic Technique (Group III)
0	6	22	25
>0 ≤ 2	24	9	10
>2 ≤ 3	3	6	4
>3 ≤ 5	4	1	5
>5	3	2	1

*The values are given as the number of patients.

100 points on the Lysholm and Gillquist scale^{26,39}. The pivot-shift test was positive for these two patients, as well as for the four in whom the graft had failed.

Over-all, the side-to-side difference that was found on testing at maximum manual force was three millimeters or less in thirty-three patients (83 per cent) in group I, thirty-seven patients (93 per cent) in group II, and thirty-nine patients (87 per cent) in group III (Table II); the findings in group II were significantly different from those in groups I and III ($p < 0.08$, Student *t* test). The side-to-side difference was two millimeters or less in thirty patients (75 per cent) in group I, thirty-one patients (78 per cent) in group II, and thirty-five patients (78 per cent) in group III (Table II); these findings were not significantly different among the groups ($p > 0.1$). Without exception, the greatest differences were mea-

sured on testing at maximum manual force and are thus reported for the purposes of this study.

The side-to-side difference on the Lachman test was graded as 0 in twenty-six patients (65 per cent) in group I, twenty-four patients (60 per cent) in group II, and thirty-four patients (76 per cent) in group III. The difference was grade 1 or less in thirty-seven patients (93 per cent) in group I, thirty-seven patients (93 per cent) in group II, and thirty-nine patients (87 per cent) in group III.

Testing with the KT-2000 arthrometer during the follow-up examinations showed no evidence of increasing laxity over time in any of the groups.

Range of Motion

Two knees in group I, six in group II, and two in group III lost terminal flexion (average, 11 degrees; range, 7 to 18 degrees). Only two knees (in group II) lost extension: one lost 2 degrees and the other, 4 degrees. No technical factors were associated with a loss of motion. Specifically, the placement of the tunnel, as identified on the radiographs that were made at the time of the most recent follow-up, was not associated with a loss of motion for any of the knees.

Patellar Crepitus

Patellar crepitus in the involved knee was subjectively compared with that in the contralateral knee. In group I, two knees had moderate crepitus and two had mild crepitus; in group II, two knees had moderate crep-



FIG. 3-B



FIG. 3-C

Anteroposterior and lateral radiographs of the knee, made at thirty-six months. The patient was a nineteen-year-old man who played collegiate football as a linebacker.

itus; and in group III, one knee had moderate crepitus and four had trace crepitus.

Deficit in Strength of the Quadriceps and Hamstrings

On isokinetic testing at all of the speeds evaluated, the deficit in the strength of the quadriceps was at least 10 per cent in nine patients in group I, eleven patients in group II, and eighteen patients in group III (Table III). Five of the patients in group III had a deficit of at least 20 per cent, but no patient in group I or II had a deficit of this magnitude. The groups did not differ significantly with regard to the deficit in the strength of the quadriceps.

None of the patients in group I or II had a deficit in the strength of the hamstrings that was 20 per cent or more; none of the patients in group III had a deficit of the hamstrings at any testing speed (Table III). There was a deficit of 10 per cent or more (but less than 20 per cent) in eight patients in group I and five patients in group II.

Serial data that were obtained at the two-year follow-up examination and annually thereafter showed that the muscular deficits diminished with time.

Subsequent Operative Procedures

Six additional operations were done in four patients in group I. As previously noted, two patients had a second reconstruction of the anterior cruciate ligament; one of them also had an excision of a painful plica and of parapatellar adhesions. Two other patients had a partial lateral meniscectomy and a lateral retinacular release.

Nine additional operative procedures were done in six patients in group II. Two patients had a second reconstruction of the anterior cruciate ligament; one of them also had partial medial and lateral meniscectomies and the other had a partial medial meniscectomy. One patient had a repair of the medial meniscus and removal of a loose osteochondral body. Three patients had removal of deep hardware; one of the patients had a revision of the scar from the initial operation, and one had a manipulation under anesthesia to improve motion. The third patient had subsequent removal of hardware, two manipulations under anesthesia to improve motion, and arthroscopic lysis of adhesions.

Four patients in group III had a subsequent operation. One patient had arthroscopic removal of a loose metallic foreign body in the knee, one had repair of the medial meniscus, one had débridement of adhesions on the intercondylar notch and a lateral retinacular release, and one had removal of deep hardware.

Significantly fewer additional procedures were done in group III ($p < 0.08$, Student t test, greater than 90 per cent confidence level). When the patients who had had a second reconstruction of the anterior cruciate ligament were excluded, the only patients whose outcome was affected by the additional operation were those who had had a lateral retinacular release (two in

TABLE III
POSTOPERATIVE STRENGTH DEFICITS ON ISOKINETIC TESTING

Deficit (Per cent)	Semitendinosus- Gracilis Graft (Group I)*	Patellar Ligament Graft*	
		Two-Incision Technique (Group II)	Endoscopic Technique (Group III)
Quadriceps			
≥10	9	11	18
≥20	0	0	5
Hamstrings			
≥10	8	5	0
≥20	0	0	0

*The values are given as the number of knees.

group I and one in group III). For all three patients, the knee was rated B on the International Knee Documentation Committee scale at the most recent follow-up examination.

One of the four patients who had a second reconstruction of the anterior cruciate ligament was followed for three years after the second procedure; the knee was rated B because of a side-to-side difference of four millimeters on testing at maximum manual force with the KT-2000 arthrometer. At the time of the latest follow-up, the patient was playing collegiate football. The other three patients had a side-to-side difference of three millimeters or less but were not followed for the minimum of twenty-four months after the second procedure.

Findings on Postoperative Radiographs

At the follow-up evaluations, an anteroposterior radiograph was made with the knee in full extension and a lateral radiograph was made with the knee in 40 degrees of flexion. There was no attempt to make reproducible radiographs to determine if there was divergence of the interference screw. There was no radiographic evidence, in any group, of progressive degenerative changes in the knee during the interval of this study, despite the substantial number of meniscal injuries in the study population.

The failure of one graft in group I and one in group III was thought to be related to the anterior placement of the tibial tunnel, as seen on the lateral radiograph. Impingement of the graft in the femoral intercondylar notch during terminal extension of the knee probably resulted from the anterior placement and ultimately led to failure of the graft.

Functional Testing

According to the system of Lysholm and Gillquist^{26,39}, the score was 90 points or more for thirty-five knees (88 per cent) in group I, thirty-six knees (90 per cent) in group II, and forty-two knees (93 per cent) in group III. The value on the one-leg-hop test⁸ was at least 90 per cent that for the contralateral limb in thirty-six patients (90 per cent) in group I, thirty-seven (93 per cent) in

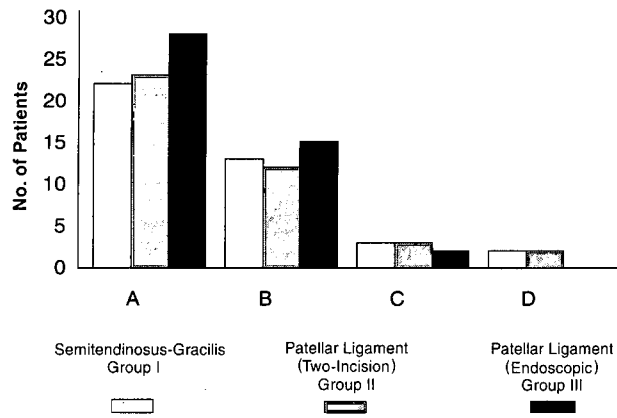


FIG. 4

Ratings for the knees according to the International Knee Documentation Committee scale¹⁷.

group II, and forty (89 per cent) in group III.

The International Knee Documentation Committee scale¹⁷ proved to be more discriminating. According to this scale, a knee with normal function is rated A. The grade becomes lower on the basis of the patient's subjective assessment, symptoms, loss of motion, laxity of three millimeters or more on instrumented testing, pivot shift, crepitus, narrowing of the cartilage space on radiographs, and a one-leg-hop value of less than 90 per cent that for the contralateral side. A knee that has nearly normal function and three to five millimeters of laxity is graded B. If there is six to ten millimeters of laxity, the grade is C (abnormal). Any crepitus results in a grade of C or D.

In group I, the rating was A for twenty-two knees (55 per cent), B for thirteen (33 per cent), C (because of laxity or crepitus) for three (8 per cent), and D (because the graft failed) for two (5 per cent) (Fig. 4).

Twenty-three knees (58 per cent) in group II were rated A, twelve (30 per cent) were rated B, three (8 per cent) were rated C (for crepitus or loss of flexion), and two (5 per cent) were rated D (because of failure of the graft) (Fig. 4).

In group III, twenty-eight knees (62 per cent) were rated A, fifteen (33 per cent) were rated B, and two (4 per cent) were rated C (for crepitus or laxity). No knee had a rating of D (Fig. 4).

In summary, 113 (90 per cent) of the 125 knees had either normal function (grade A) or nearly normal function (grade B). The only significant finding was that no knee in group III was rated D ($p < 0.002$, 94 per cent confidence level).

Discussion

The results of reconstruction of the anterior cruciate ligament with the three techniques used in this series were consistent with those that have been reported previously^{1,3,6,14,15,18,19,21,27,30,34,42}.

Although 90 per cent (113) of all of the knees in this series had normal or nearly normal function, regardless

of the technique that had been used, this prospective study illustrates several points.

With the numbers available, there was no significant difference in outcome between the acute reconstructions (those performed less than three weeks after the injury) and the late ones ($p > 0.1$), regardless of the technique used. The four failed grafts (two in group I and two in group II) were in knees that had had a reconstruction in the acute period. The patient who had a side-to-side difference of nine millimeters on testing with the KT-2000 arthrometer had also had the operation done acutely. In contrast, the patient in group III who had a side-to-side difference of seven millimeters had the reconstruction done after the knee had been chronically unstable for three years.

It has been reported that reconstruction of the anterior cruciate ligament with a semitendinosus-gracilis graft is successful when it is performed acutely but is associated with an unacceptable rate of failure when it is used to reconstruct a chronically unstable knee^{1,20,33,35,40}. The results of this study, however, are in agreement with those of Karlson et al.²¹, which showed equal success after acute and late reconstructions with this graft. The results of the current prospective study are similar to those of the prospective study by Marder et al.²⁷. They found no difference in either functional outcome or laxity on instrumented testing when arthroscopically assisted reconstruction with a patellar ligament graft was compared with that performed with a semitendinosus-gracilis graft.

Fixation of an autogenous graft has been a concern of previous authors^{16,20-23,30,32,35} as well as a concern of mine. However, to eliminate variables, no change in the postoperative management was made to accommodate any specific technique of reconstruction. The resumption of an immediate full range of knee motion and full weight-bearing was emphasized. The results of the present study confirm the finding of Steiner et al.³⁸ that good fixation of a semitendinosus-gracilis graft can be achieved. With the numbers available, no significant difference among the three groups could be detected with regard to side-to-side differences of two millimeters or less on testing with the KT-2000 arthrometer ($p > 0.1$). Additionally, concerns have been raised about the greater divergence of the femoral interference screw when it has been placed with the endoscopic technique and the possible long-term consequences of this divergence on the fixation of an autogenous patellar ligament graft^{19,25}. This concern was not borne out in this study; measurements of laxity were not significantly different between groups II and III ($p > 0.1$). This supports the findings of Garfinkel et al.¹⁴ and those of Sgaglione and Schwartz³⁴, who reported no difference in outcome between the two-incision and the endoscopic technique when a patellar ligament graft was used.

The infrequency of pain in the anterior part of the knee and of an increase in patellofemoral crepitus in all

groups in this study was thought to be due to the encouragement of full weight-bearing and a full range of motion on the morning after the operation. The decreased prevalence of pain in the anterior part of the knee is evidence that supports the rehabilitation studies of Shelbourne and Nitz³⁶ and as well as of Sachs et al.³³. Sachs et al. reported that a flexion contracture of 5 degrees or more corresponds with patellofemoral pain. No patient in the present study had a flexion contracture of 5 degrees or more.

Surprisingly, advancing age had no bearing on the outcome. Ratings of C and D on the scale of the International Knee Documentation Committee¹⁷ and scores of less than 90 points according to the system of Lysholm and Gillquist^{26,39} were distributed equally among all age-groups and between men and women. Age was not found to be a factor in any loss of knee flexion.

It is disturbing that the only deficits on isokinetic testing that were 20 per cent or more were found in group III. Although, with the numbers available, no difference between group III and the other groups could be determined ($p > 0.1$), five patients (11 per cent) in group III had a deficit in the strength of the quadriceps that was 20 per cent or more. No significant difference in this deficit could be found between groups I and II. In contrast to the findings of Marder et al.²⁷, no patient in any of the three groups had a deficit in the strength of the hamstrings of 20 per cent or more. No patient in group III had a deficit of 10 per cent or more. In groups I and II, in which the operative technique involved a rear-entry incision in the lateral part of the thigh, thirteen patients (eight [20 per cent] in group

I and five [13 per cent] in group II) had a deficit of 10 per cent or more (but less than 20 per cent). These deficits were thought to relate to the posterolateral dissection rather than to the choice of the donor site of the autogenous graft.

Forty-three (96 per cent) of the forty-five knees in group III were rated as having normal or nearly normal function according to the scale of the International Knee Documentation Committee¹⁷, compared with seventy (88 per cent) of the eighty knees in the other two groups; however, this difference was not significant ($p > 0.1$).

In conclusion, arthroscopically assisted reconstruction of the anterior cruciate ligament with use of either an autogenous semitendinosus-gracilis or patellar ligament graft can be expected to produce good results, regardless of how long the knee has been unstable. The results of the current study show no clear advantage of a patellar ligament graft compared with a semitendinosus-gracilis graft or of the endoscopic technique compared with the two-incision technique when a patellar ligament graft is used.

The prospective randomized study of arthroscopically assisted reconstructions of the anterior cruciate ligament in these 125 patients is continuing, and an additional 100 to 125 patients are also being studied. It is hoped that these studies will delineate more clearly the effect of subtle differences among reconstructive techniques on the outcome after follow-up of longer duration.

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